Global Angular Blocks for the Middle-Ratio Shell Proxy

Global analytic angular module

Abstract

The global structural reduction for the spherical-shell proxy leaves an angular rounding defect $\mathcal{E}_{\mathrm{ang}}$ and a smooth positive payment $\mathcal{H}$. This note controls the defect for every $0 < \rho < 1$ and $K > 0$ with at least one active radial layer. First, an exact Stieltjes formula is given with the strict common-jump convention made explicit. The negative mean of every complete angular cell is combined with the single-peaked shape of $q = -A'$ to obtain continuous block estimates. A paired radial block argument then controls the full shifted sample sum, including all common jumps, by

$$\mathcal{E}_{\mathrm{ang}} < \frac{(1-\rho^2)K^2}{6} - \frac{N}{2},$$

where N is the number of active shifted radial layers. On the upward-closed middle-ratio region $\rho_c \leq \rho \leq 39/50,\ K \geq k_{11}(\rho)$, one has $N \geq 1$. The global middle-ratio theorem is therefore reduced to the smooth scalar inequality (28), which is established in the companion scalar-positivity module.

1 Definitions and the target payment

For $a > 0$ and $z \geq 0$, define

$$G_a(z) = \begin{cases} \dfrac{1}{\pi}\left(\sqrt{a^2 - z^2} - z \arccos \dfrac{z}{a}\right), & 0 \leq z < a, \\ 0, & z \geq a. \end{cases} \tag{1}$$

Fix $0 < \rho < 1$ and $K > 0$, and put

$$A(z) := G_K(z) - G_{\rho K}(z), \qquad T := A(0) = \frac{(1-\rho)K}{\pi}. \tag{2}$$

The function A decreases strictly from T to 0. Denote its decreasing inverse by $R : [0, T] \to [0, K]$, and set

$$F(t) := R(t)^2, \qquad q(z) := -A'(z), \qquad \tau := A(\rho K) = K G_1(\rho). \tag{3}$$

Direct differentiation gives

$$q(z) = \frac{1}{\pi} \begin{cases} \arccos \dfrac{z}{K} - \arccos \dfrac{z}{\rho K}, & 0 < z < \rho K, \\ \arccos \dfrac{z}{K}, & \rho K \leq z < K. \end{cases} \tag{4}$$

Consequently

$$0 \leq q(z) \leq \frac{1}{2}, \qquad q \text{ increases on } (0, \rho K), \qquad q \text{ decreases on } (\rho K, K). \tag{5}$$

Let

$$x_n := n - \frac{1}{4}, \qquad M(r) := \max\left\{0, \left\lceil r - \frac{1}{2} \right\rceil\right\}, \qquad N := \#\{n \geq 1 : x_n < T\}. \tag{6}$$

For $1 \leq n \leq N$, write

$$r_n := R(x_n), \qquad m_n := M(r_n), \tag{7}$$

and define the exact angular rounding defect

$$\mathcal{E}_{\mathrm{ang}}(\rho, K) := \sum_{n=1}^{N} (m_n^2 - r_n^2). \tag{8}$$

The strict convention is built into M: at $r = m + 1/2$ one has $M(r) = m$, not $m + 1$.

The smooth payment obtained from the retained radial Stieltjes remainder is

$$\begin{aligned}\mathcal{H}(\rho, K) := & \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} \\ & + \frac{2\pi\rho K}{\arccos \rho}\left(\frac{\tau}{4(1+2\tau)} + \frac{3}{32}\right) \\ & + \int_{\rho K}^{K} 2z \, \frac{A(z)(1+A(z))}{(1+2A(z))^2}\, dz.\end{aligned} \tag{9}$$

The preceding structural reduction proves

$$\mathcal{E}_{\mathrm{ang}} < \mathcal{H} \quad \Longrightarrow \quad P(\rho, K) < W(\rho, K). \tag{10}$$

This note derives a smooth upper bound for the left side.

2 Complete angular cells and the single peak

Define the angular sawtooth error

$$e(r) := M(r)^2 - r^2. \tag{11}$$

For $m \geq 1$, the strict m-cell is

$$I_m = (m - 1/2, m + 1/2], \qquad e(r) = m^2 - r^2 \quad (r \in I_m).$$

Endpoint values do not affect the following Lebesgue integrals. The positive and negative half-cell masses are

$$P_m := \int_{m-1/2}^{m} (m^2 - r^2)\, dr = \frac{m}{4} - \frac{1}{24}, \tag{12}$$

$$Q_m := -\int_{m}^{m+1/2} (m^2 - r^2)\, dr = \frac{m}{4} + \frac{1}{24}. \tag{13}$$

In particular,

$$\int_{m-1/2}^{m+1/2} e(r)\, dr = P_m - Q_m = -\frac{1}{12}. \tag{14}$$

Lemma 1 (Weighted complete-cell bounds)**.** *Let $m \geq 1$ and suppose the complete cell $[m-1/2, m+1/2]$ lies in $[0, K]$.*

1. If $m + 1/2 \le \rho K$, then

$$\int_{m-1/2}^{m+1/2} e(r)q(r)\,dr \le -\frac{q(m)}{12} \le 0. \tag{15}$$

2. If $m - 1/2 \ge \rho K$, then

$$\int_{m-1/2}^{m+1/2} e(r)q(r)\,dr \le P_m q(m-1/2) - Q_m q(m+1/2). \tag{16}$$

3. On the unique cell, if any, whose interior contains ρK,

$$\int_{m-1/2}^{m+1/2} e(r)q(r)\,dr \le P_m q(\rho K). \tag{17}$$

The initial partial cell $0 \le r \le 1/2$ contributes nonpositively.

Proof. On the left half of a cell $e \ge 0$, and on the right half $e \le 0$. If q is increasing, then $q(r) \le q(m)$ on the positive half and $q(r) \ge q(m)$ on the negative half. Equations (12)–(14) give (15).

If q is decreasing, then $q(r) \le q(m-1/2)$ on the positive half and $q(r) \ge q(m+1/2)$ on the negative half. This gives (16). On the peak cell, discard the negative half and use $q \le q(\rho K)$ on the positive half. Finally, $M(r) = 0$ and $e(r) = -r^2$ for $0 \le r \le 1/2$. □

Thus the continuous angular defect has a favorable analytic block structure:

$$\int_0^T e(R(t))\,dt = \int_0^K e(r)q(r)\,dr. \tag{18}$$

All complete cells below the peak are nonpositive, and the decreasing side is controlled by the endpoint differences in (16). The difficulty is passing from the integral in (18) to the shifted samples in (8); the next subsection records that passage with all common jumps included.

3 Exact common-jump convention

Let

$$h(t) := e(R(t)), \qquad C(t) := \#\{n : 0 < x_n \le t,\ x_n < T\}, \qquad p(t) := C(t) - t. \tag{19}$$

Choose the right-continuous versions of C, p, and h. Right continuity of h is exactly the strict angular convention: when $R(t) = m + 1/2$, the post-jump value uses $M = m$. The atomic measure dC therefore samples the correct value even when a radial point x_n is also an angular wall.

Lemma 2 (Euler identity with common jumps)**.** *With the preceding representatives,*

$$\mathcal{E}_{\text{ang}} = \int_0^T h(t)\,dt - \int_{(0,T]} p(t^-)\,dh(t). \tag{20}$$

At a common radial–angular wall the second integral uses the left value $p(t^-) = -3/4$; no extra or omitted wall term is permitted.

Proof. The measure dC has a unit atom at every active x_n, so

$$\mathcal{E}_{\text{ang}} = \int_{(0,T)} h(t)\, dC(t).$$

Since $dC = dt + dp$,

$$\mathcal{E}_{\text{ang}} = \int_0^T h(t)\, dt + \int_{(0,T]} h(t)\, dp(t).$$

For right-continuous functions of bounded variation, the product formula in the form

$$\int_{(0,T]} h(t)\, dp(t) = [hp]_0^T - \int_{(0,T]} p(t^-)\, dh(t)$$

already contains the product of simultaneous jumps. Here $p(0) = 0$ and $h(T) = e(0) = 0$, so the boundary term vanishes. This proves (20). If $t = x_n$, then $C(t^-) = n-1$ and hence $p(t^-) = n - 1 - (n - 1/4) = -3/4$, including a common angular wall. □

Equation (20) explains why the negative means in (14) cannot be used without a jump audit. The second term contains both the continuous rise inside each angular cell and the downward atom at its strict lower wall. A direct sign deletion would be invalid. The paired estimate below controls the integral and all its atoms simultaneously.

4 A paired radial-block estimate

The cap $q \le 1/2$ in (5) forces a useful separation: one unit of radial action spans at least two units of angular radius. The shift $x_n = n - 1/4$ leaves a left block of length $3/4$, enough to pay the worst angular rounding in that layer.

Lemma 3 (One-layer paired block)**.** *For every active n,*

$$r_n \le \frac{4}{3}\int_{n-1}^{x_n} R(t)\, dt - \frac{3}{4}, \tag{21}$$

and

$$m_n^2 - r_n^2 < r_n + \frac{1}{4}. \tag{22}$$

Both statements remain valid, with the displayed strictness, at every radial–angular common wall.

Proof. Fix $t \in [n-1, x_n]$ and put $z = R(t) \ge r_n$. Since $A(r_n) = x_n$, (5) gives

$$x_n - t = A(r_n) - A(z) = \int_{r_n}^{z} q(s)\, ds \le \frac{z - r_n}{2}.$$

Hence

$$R(t) \ge r_n + 2(x_n - t).$$

Integration over the interval of length $3/4$ yields

$$\int_{n-1}^{x_n} R(t)\, dt \ge \frac{3}{4} r_n + \frac{9}{16},$$

which is (21).

By strict counting, $m_n < r_n + 1/2$ even when r_n is a half-integer: at $r_n = m + 1/2$ the value is $m_n = m$. Squaring gives

$$m_n^2 < (r_n + 1/2)^2 = r_n^2 + r_n + 1/4,$$

and proves (22). □

Proposition 4 (Global angular block bound)**.** *For every* $0 < \rho < 1$ *and every* $K > 0$ *with* $N \geq 1$*,*

$$\boxed{\mathcal{E}_{\text{ang}}(\rho, K) < \frac{(1-\rho^2)K^2}{6} - \frac{N}{2}.} \tag{23}$$

In particular,

$$\mathcal{E}_{\text{ang}}(\rho, K) < \frac{(1-\rho^2)K^2}{6} - \frac{1}{2}. \tag{24}$$

Proof. The left blocks $[n-1, x_n]$, $1 \leq n \leq N$, are disjoint subsets of $[0, T]$. Summing (21) and then (22) gives

$$\begin{aligned}
\mathcal{E}_{\text{ang}} &< \sum_{n=1}^{N} \left(r_n + \frac{1}{4} \right) \\
&\leq \frac{4}{3} \sum_{n=1}^{N} \int_{n-1}^{x_n} R(t)\, dt - \frac{N}{2} \\
&\leq \frac{4}{3} \int_0^T R(t)\, dt - \frac{N}{2}.
\end{aligned}$$

The area identity for decreasing inverses gives

$$\int_0^T R(t)\, dt = \int_0^K A(z)\, dz.$$

Scaling $z = as$ in (1) and using

$$\int_0^1 \sqrt{1-s^2}\, ds = \frac{\pi}{4}, \qquad \int_0^1 s \arccos s\, ds = \frac{\pi}{8},$$

shows that

$$\int_0^a G_a(z)\, dz = \frac{a^2}{8}.$$

Consequently

$$\int_0^T R(t)\, dt = \frac{(1-\rho^2)K^2}{8},$$

which proves (23). Equation (24) follows from $N \geq 1$. $\square$

5 The upward-closed middle-ratio scalar inequality

Set

$$\rho_c := \frac{1}{1+2\pi}, \qquad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}, \tag{25}$$

and consider the upward-closed middle-ratio region

$$\mathcal{R}_{\text{c}} := \left\{ (\rho, K) : \rho_c \leq \rho \leq \frac{39}{50}, \quad K \geq k_{11}(\rho) \right\}. \tag{26}$$

On this set,

$$T \geq \frac{(1-\rho)k_{11}(\rho)}{\pi} = \sqrt{1 + \frac{132(1-\rho)^2}{\pi^2}} > 1, \tag{27}$$

so $N \geq 1$. Proposition 4 and (10) reduce the global middle-ratio theorem to the single inequality

$$\boxed{\mathcal{S}(\rho, K) := \mathcal{H}(\rho, K) - \frac{(1-\rho^2)K^2}{6} + \frac{1}{2} > 0} \qquad ((\rho, K) \in \mathcal{R}_{\mathrm{c}}). \tag{28}$$

Everything in $\mathcal{S}$ is smooth at every finite point of $\mathcal{R}_{\mathrm{c}}$; the floor functions and all radial–angular common jumps have disappeared. Substituting (9) gives the explicit form

$$\begin{aligned} \mathcal{S}(\rho, K) = {} & \frac{(5\rho^2 - 2)K^2}{12} - \frac{\pi \rho K}{4(1-\rho)} + \frac{1}{2} \\ & + \frac{2\pi\rho K}{\arccos \rho} \left(\frac{K G_1(\rho)}{4(1 + 2K G_1(\rho))} + \frac{3}{32} \right) \\ & + \int_{\rho K}^{K} 2z \, \frac{A(z)(1 + A(z))}{(1 + 2A(z))^2} \, dz. \end{aligned} \tag{29}$$

The elementary identity

$$\frac{u(1+u)}{(1+2u)^2} = \frac{1}{4} - \frac{1}{4(1+2u)^2} \tag{30}$$

turns the same scalar gap into the more useful loss form

$$\begin{aligned} \mathcal{S}(\rho, K) = {} & \frac{(1 + 2\rho^2)K^2}{12} - \frac{1}{2} \int_{\rho K}^{K} \frac{z}{(1 + 2A(z))^2} \, dz \\ & - \frac{\pi \rho K}{4(1-\rho)} + \frac{1}{2} + \frac{2\pi\rho K}{\arccos \rho} \left(\frac{K G_1(\rho)}{4(1 + 2K G_1(\rho))} + \frac{3}{32} \right). \end{aligned} \tag{31}$$

Thus the only non-endpoint quantity still requiring control is the positive boundary-layer loss integral in (31).

Remark 5 (Status and next step). The complete-cell bounds, the common-jump formula, and the global estimate (23) are proved analytically. The first remaining step within this module is the smooth scalar inequality (28). The companion scalar-positivity module proves it on all of $\mathcal{R}_{\mathrm{c}}$ by a uniform upper bound for the boundary-layer loss in (31) and strict positivity on the base curve. No upper cutoff in K occurs in the angular estimate or in the verification that $N \geq 1$.